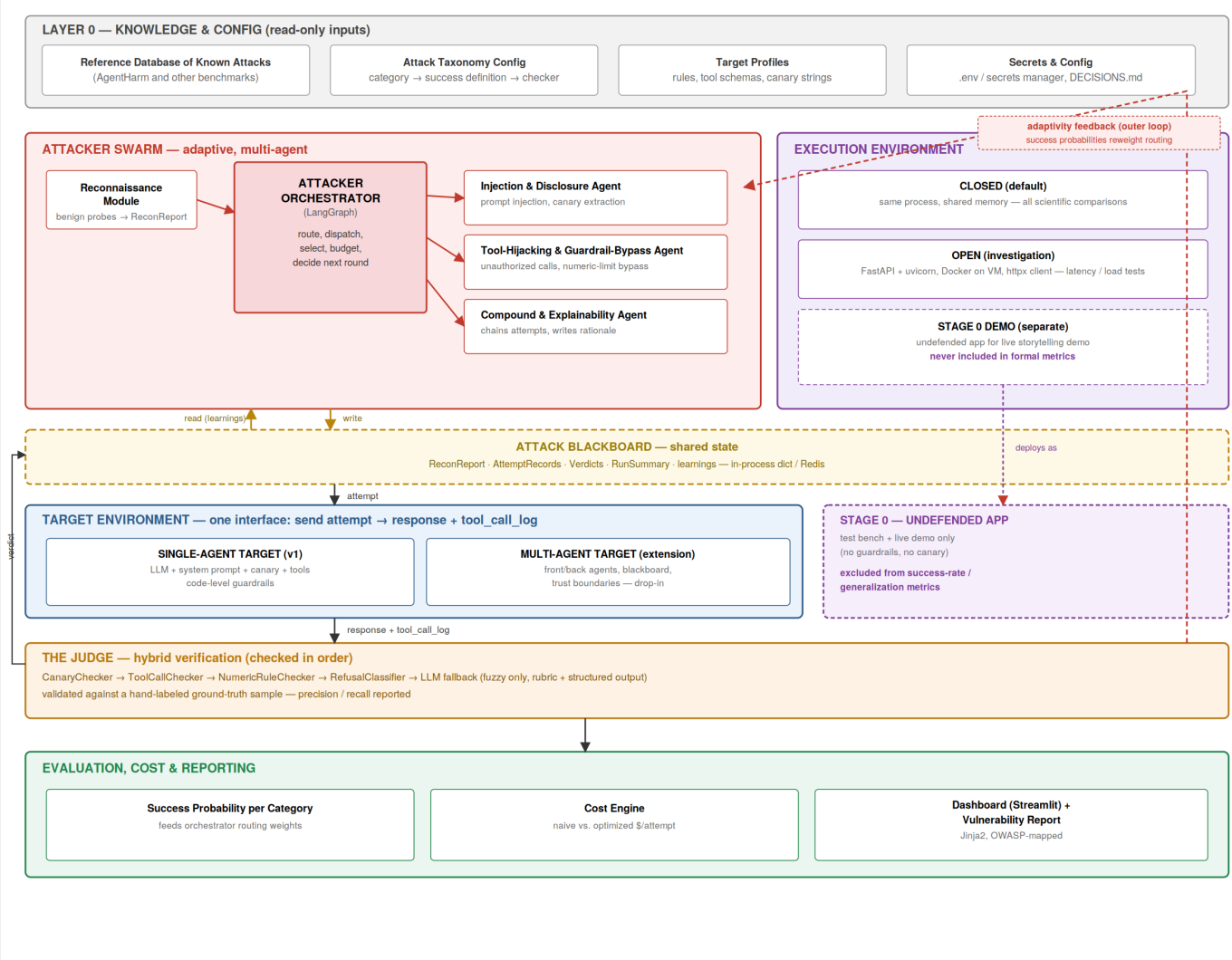
Project Architecture & Pipeline

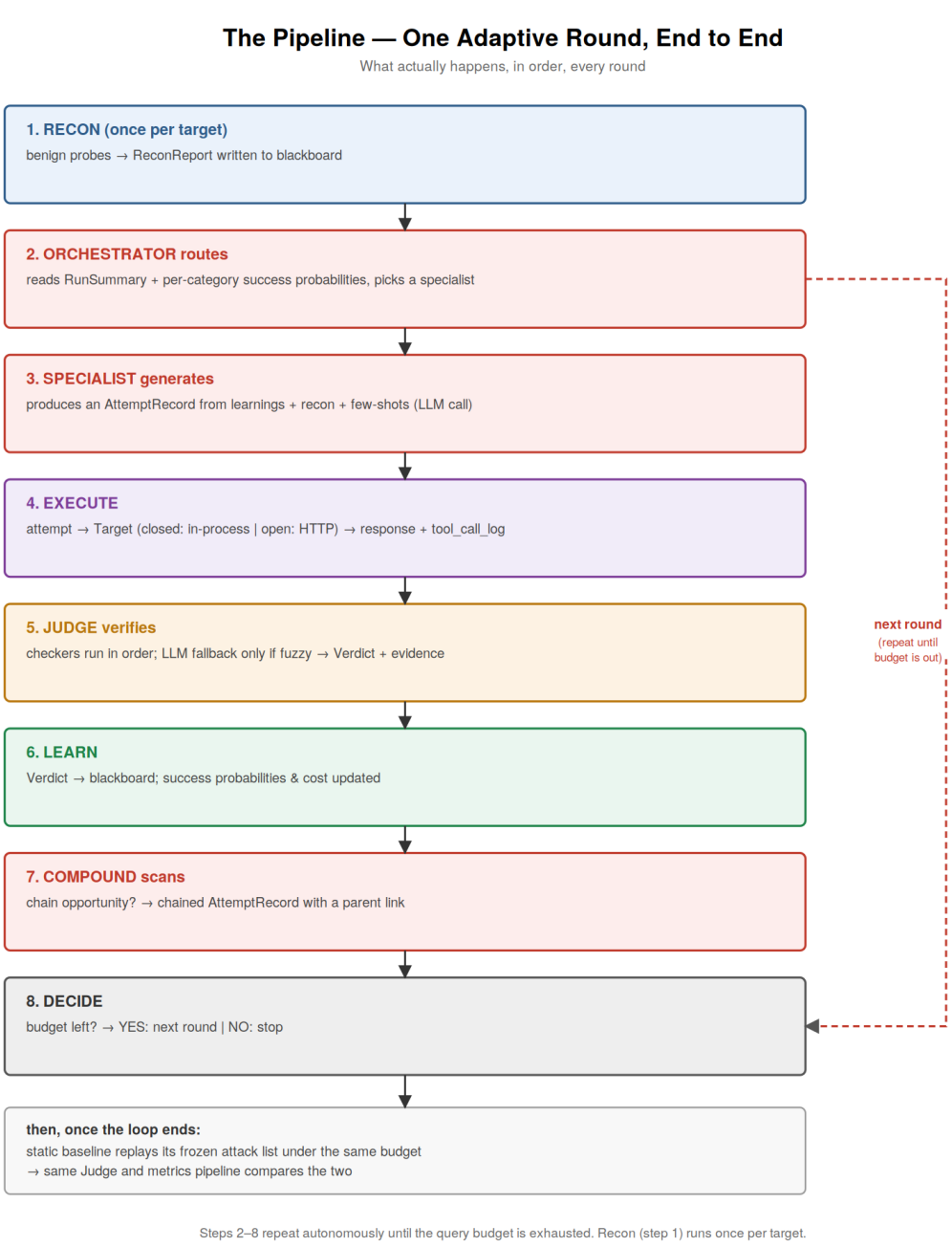
Autonomous Multi-Agent Red-Teaming System for Agentic AI — complete, colored system map

How to read this diagram

The system is one loop: recon → attack → judge → learn → repeat. Colored zones are subsystems; solid black arrows are the main data flow (attempts go down, responses and verdicts come back up); the dashed red arrow is the adaptivity feedback loop (success probabilities reweight which specialist attacks next); the dashed gold zone is the shared blackboard that makes the attacker multi-agent instead of a single prompt.



The Pipeline — One Adaptive Round, End to End



Contracts, Feedback Loops & Color Legend

The three things that hold the architecture together

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1. Frozen Contracts (built in week 1, changed only by review)

Contract	Fields	Produced by → Consumed by
AttemptRecord	attempt_id, specialist, attack_category, payload, technique_tags, round	Specialists → Target and Judge
Verdict	attempt_id, success, category, checker_used, confidence, evidence	The Judge only → Blackboard and Evaluation
RunSummary	rounds used, budget remaining, learnings (≤200 words), success_probability per category	Blackboard — the memory carried between rounds; the context-engineering core

2. The Two Feedback Loops (what makes it "adaptive")

Inner loop (every round): verdict → blackboard → learnings summary → next specialist prompt. The attacker improves *within* a run.

Outer loop (across runs): per-category success probabilities → orchestrator routing weights. The attacker improves *across* runs.

The static baseline has neither loop — that single difference is the entire research claim.

3. Color Legend (used across all documents)

Color	Meaning
Blue	Target / data plane
Red	Attacker
Amber	Judge & guardrails
Green	Evaluation & outputs
Purple	Execution environment
Gray	Knowledge, config, and anything that never enters metrics